

Data types

Individual data (0's and 1's):

- For each individual, know the initial size and whether it was alive or dead after some time interval

Summarized data (prop. surviving):

- Know the initial number of individuals in some size range and the proportion that survived some time interval

Data integration: option 1 (current)

Individual data (0's and 1's)

Summarized data (prop. surviving)

use size ranges and prop.
survival to randomly draw
individual sizes and survivals

Estimated individual data (0's and 1's) with
multiple random replicates

Full data (0's and 1's)

Randomly draw n individuals from each size
class and calculate proportion that survived;
repeat 100 times

Parameter estimates

Main issues: 1) for cases where size ranges in the proportion data span multiple model size classes, size-survival pairings are uncertain, 2) sample sizes: draw n individuals to estimate survival parameter, but sample size in the model varies (e.g., $n = 20$ is used for parameter estimate, but there are often >1000 babies each year in model simulation)

Data integration: option 2

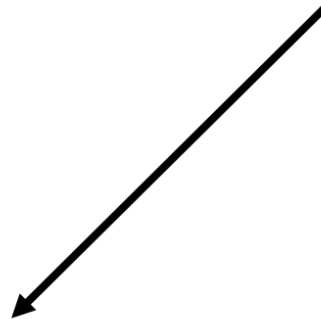
Individual data (0's and 1's)



Proportions surviving in each
location/year/size class



Summarized data (prop. surviving)



Full data (proportions surviving)



Randomly draw rows to generate parameter
sets

Parameter estimates

Main issues: the sample sizes that underlie the proportional survival values can vary greatly across studies/size classes